

Course Name: Computer vision with OpenCV.

Code: PSA02

CO2: Apply image enhancement and segmentation technique

Assignment tool weightage: low

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1. As an image appears too bright, explain and justify in most suitable grey level to improve visibility, clearly describing the concept.

The most suitable to apply an algorithmic transformation which compress high-intensity values while expanding darker ones.

$$S = C \cdot \log(1+r).$$

Mapping a narrow range of bright pixel values into a wider displayable range.

2. A dark image lacks detail. Describe and justify an appropriate grey level transformation to enhance visibility.

use a power-law (gamma) transformation with.

$$r < 1 \quad (S = C \cdot r^{\gamma})$$

This stretches low-intensity values over a wider

range, brightening shadow regions without over-
bright areas. Hidden details in the dark-regions
visible.

3 An image has poor contrast. Explain a suitable
histogram-based technique for the contrast enhancement.

Apply the histogram equalisation, which redistributes
pixel intensities using the cumulative distribution
function (CDF). to spread them more uniformly across
the range. Densely packed intensity values get
stretched apart. The image becomes sharper and
more detailed.

6 A noisy image contains random intensity
variations. Explain and justify the most suitable
smoothing filter.

use a Gaussian smoothing filter which
convolves the image with a Gaussian kernel
giving higher weight to central pixels. This
averages out random noise while preserving
structure better than a simple mean filter.

edges remain relatively intact.

8 An image appears blurred after smoothing. Explain
and justify a suitable sharpening method.

Apply unsharp masking, which highlights high-

frequency edge components by subtracting a blurred version from the original (or convolving with a Laplacian kernel). Adding this back to the original restores fine details. The image looks crisper overall.

14. An image contains high-frequency noise. Explain and justify the appropriate frequency domain filter.

Use a low-pass filter (Ideal, Butterworth via Fourier transform). It attenuates high-frequency components while retaining low-frequency components. The filtered spectrum is then inverse-transformed back to the spatial domain. This produces a smoother, noise-reduced image.

15. An object is clearly brighter than the background. Explain and justify the segmentation technique used.

Apply global thresholding where a single threshold value separates pixels into the object and background. Since the intensity is high, one fixed threshold will work. It produces a clean binary segmented image.

17. An image requires identification of object edge. Explain and justify a suitable edge detection method.

Use the Canny edge detector, which applies.

Smoothing and gradient computation, non-maximum suppression and hysteresis thresholding. It reduces the false edge detection compared to simple operator. The result is thin.

19: Detected edges in an image appear broken. Explain and justify a method to improve edge continuity. Apply the morphological operations like dilation or edge linking algorithms. There are nearby broken edge segments. by filling small gaps. It ensures object boundaries from continuous closed contours. This improves subsequent shape or boundary.

27: A grayscale image shows poor separation b/w object and background. Describe and justify a thresholding approach. Use Otsu's threshold which automatically calculates an optimal threshold by maximizing b/w class variance. It analyzes the histogram instead of relying on manual selection. This results in an accurately segmented binary image.